

Project: Host scanning and analysis

Metasploitable 2 Reconnaissance & Vulnerability Analysis Authors: Andersson David Sánchez Méndez & Cristian Santiago Pedraza Rodríguez Target IP: 192.168.106.129 Attacker: 192.168.106.128 Status: Completed Description: This project contains the structured data, tool outputs, and technical findings derived from the security assessment of a Metasploitable 2 instance.

Methodology

Approach: Active Reconnaissance / Black Box. Standard: Technical Security Assessment.

- Process:
 - ◇ Network discovery and connectivity verification (Ping/Ifconfig).
 - ◇ Service enumeration and banner grabbing using Nmap.
 - ◇ Version analysis and vulnerability mapping (CVE correlation).
 - ◇ Evidence integrity verification via SHA256 hashing.

Results

Vulnerability Table

Port	Service	Version	Short Description	Severity
21	FTP	vsftpd 2.3.4	Backdoored version (CVE-2011-2523)	Critical
22	SSH	OpenSSH 4.7p1	Accessible service; outdated version	Medium
23	Telnet	Linux telnetd	Unencrypted management protocol	High
80	HTTP	Apache 2.2.8	Hosting vulnerable web applications	High
445	SMB	Samba 3.x	Potential for RCE (SambaCry/EternalRed)	Critical
3306	MySQL	5.0.51a	Database exposed via network	High

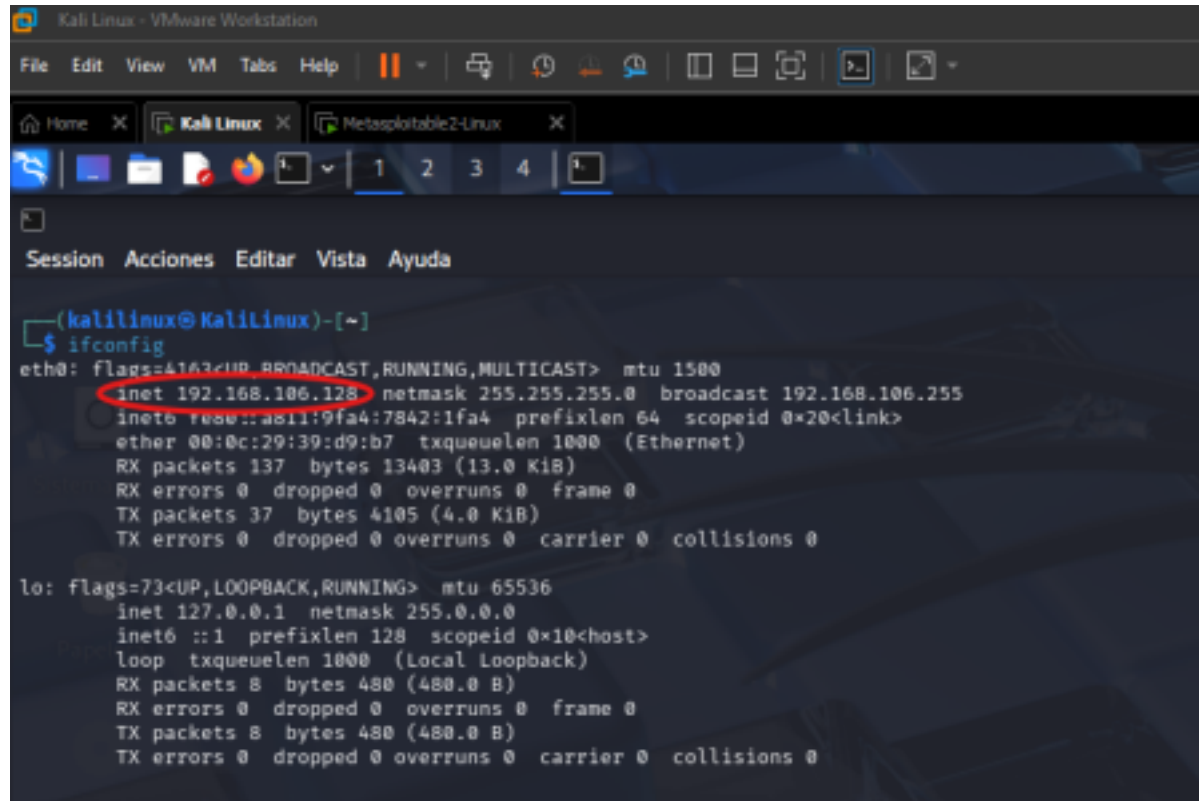
Highlighted Finding: Remote Root Compromise (vsftpd 2.3.4) Summary: The FTP service contains a backdoor

(CVE-2011-2523) that grants unauthenticated root access. Technical Description: Using a specific character sequence :) in the username triggers a shell listener on port 6200.

Evidences

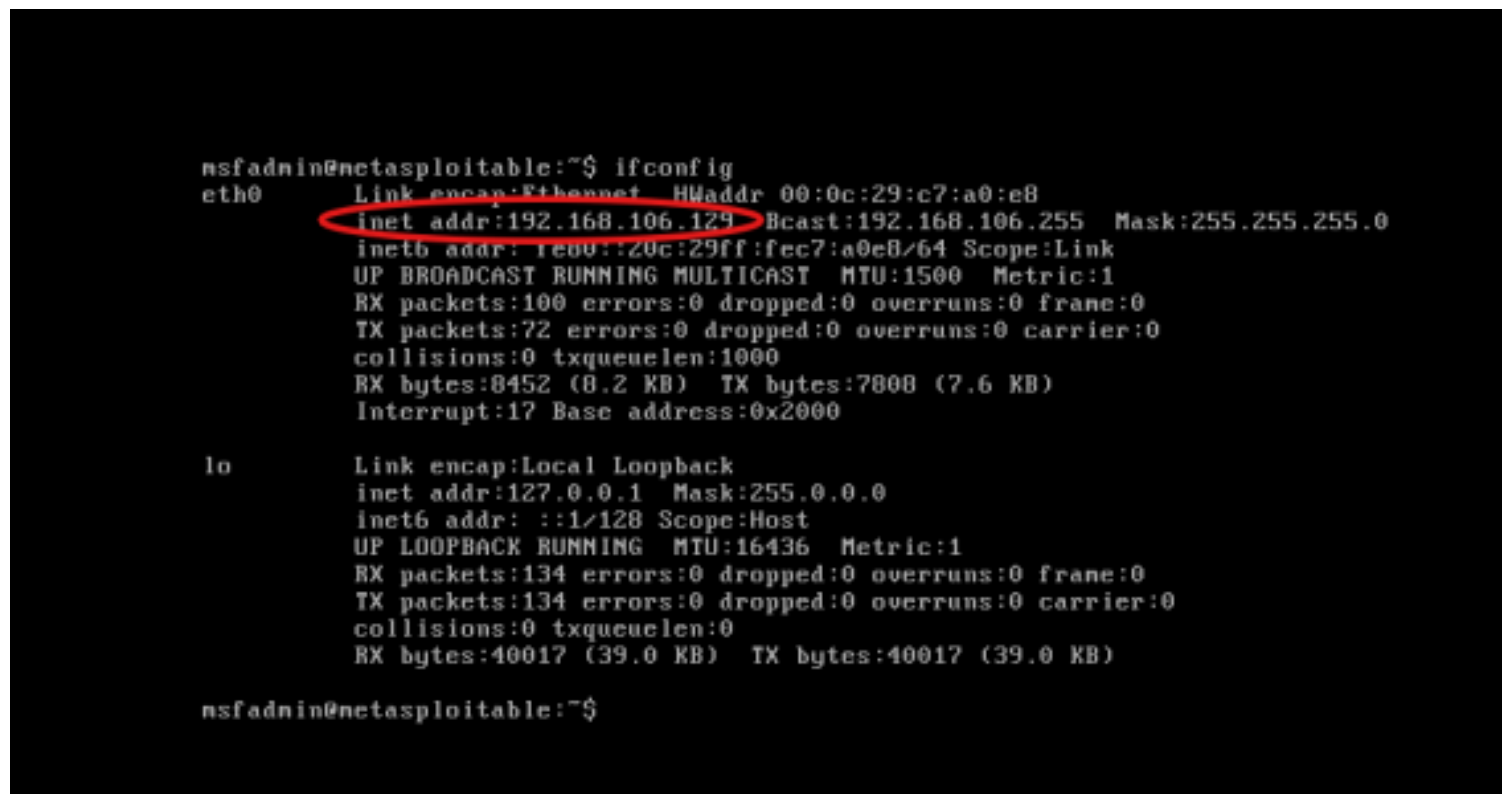
References Files & Screenshots

- ifconfig-kalilinux.png & ifconfig-metasploitable.png (Network context).



```
(kalilinux@KaliLinux)~[~]
$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.106.128 netmask 255.255.255.0 broadcast 192.168.106.255
    inet6 fe80::a511:9fa4:7842:1fa4 prefixlen 64 scopeid 0<link>
    ether 00:0c:29:39:d9:b7 txqueuelen 1000 (Ethernet)
    RX packets 137 bytes 13403 (13.0 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 37 bytes 4105 (4.0 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 8 bytes 480 (480.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 8 bytes 480 (480.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```



```
nsfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet HWaddr 00:0c:29:c7:a0:e8
          inet addr:192.168.106.129 Bcast:192.168.106.255 Mask:255.255.255.0
          inet6 addr: fe80::20c:29ff:fec7:a0e8/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
          RX packets:100 errors:0 dropped:0 overruns:0 frame:0
          TX packets:72 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:8452 (8.2 KB) TX bytes:7800 (7.6 KB)
          Interrupt:17 Base address:0x2000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1 Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING MTU:16436 Metric:1
          RX packets:134 errors:0 dropped:0 overruns:0 frame:0
          TX packets:134 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:40017 (39.0 KB) TX bytes:40017 (39.0 KB)

nsfadmin@metasploitable:~$
```

- ping-kali-to-metasploitable.png & ping-metasploitable-to-kali.png (Connectivity proof).

```
Home X Kali Linux X Metasploitable2 Linux X
1 2 3 4
Session Acciones Editar Vista Ayuda
(kalilinux@kalilinux)-[~]
$ ping 192.168.106.129
PING 192.168.106.129 (192.168.106.129) 56(84) bytes of data:
64 bytes from 192.168.106.129: icmp_seq=1 ttl=64 time=0.939 ms
64 bytes from 192.168.106.129: icmp_seq=2 ttl=64 time=1.16 ms
64 bytes from 192.168.106.129: icmp_seq=3 ttl=64 time=1.25 ms
64 bytes from 192.168.106.129: icmp_seq=4 ttl=64 time=1.81 ms
64 bytes from 192.168.106.129: icmp_seq=5 ttl=64 time=1.90 ms
64 bytes from 192.168.106.129: icmp_seq=6 ttl=64 time=1.37 ms
64 bytes from 192.168.106.129: icmp_seq=7 ttl=64 time=1.65 ms
64 bytes from 192.168.106.129: icmp_seq=8 ttl=64 time=1.22 ms
64 bytes from 192.168.106.129: icmp_seq=9 ttl=64 time=0.982 ms
64 bytes from 192.168.106.129: icmp_seq=10 ttl=64 time=0.842 ms
64 bytes from 192.168.106.129: icmp_seq=11 ttl=64 time=1.06 ms
64 bytes from 192.168.106.129: icmp_seq=12 ttl=64 time=1.82 ms
64 bytes from 192.168.106.129: icmp_seq=13 ttl=64 time=0.535 ms
64 bytes from 192.168.106.129: icmp_seq=14 ttl=64 time=1.47 ms
64 bytes from 192.168.106.129: icmp_seq=15 ttl=64 time=1.84 ms
64 bytes from 192.168.106.129: icmp_seq=16 ttl=64 time=0.800 ms
64 bytes from 192.168.106.129: icmp_seq=17 ttl=64 time=1.13 ms
^C
--- 192.168.106.129 ping statistics ---
17 packets transmitted, 17 received, 0% packet loss, time 1604ms
rtt min/avg/max/mdev = 0.535/1.280/1.903/0.399 ms
```

```
msfadmin@metasploitable:~$ ping 192.168.106.128
PING 192.168.106.128 (192.168.106.128) 56(84) bytes of data:
64 bytes from 192.168.106.128: icmp_seq=1 ttl=64 time=1.12 ms
64 bytes from 192.168.106.128: icmp_seq=2 ttl=64 time=1.34 ms
64 bytes from 192.168.106.128: icmp_seq=3 ttl=64 time=1.26 ms
64 bytes from 192.168.106.128: icmp_seq=4 ttl=64 time=0.835 ms
64 bytes from 192.168.106.128: icmp_seq=5 ttl=64 time=1.18 ms
64 bytes from 192.168.106.128: icmp_seq=6 ttl=64 time=0.957 ms
64 bytes from 192.168.106.128: icmp_seq=7 ttl=64 time=1.04 ms
64 bytes from 192.168.106.128: icmp_seq=8 ttl=64 time=1.12 ms

--- 192.168.106.128 ping statistics ---
8 packets transmitted, 8 received, 0% packet loss, time 6998ms
rtt min/avg/max/mdev = 0.835/1.108/1.340/0.156 ms
msfadmin@metasploitable:~$ _
```

- nmap-kali-to-metasploitable-part1.png to part3.png (Service enumeration evidence).


```
Session Acciones Editor Vista Ayuda
8087/tcp open  irc      UnrealIRCd
|_ irc-info:
|_ users: 1
|_ servers: 1
|_ users: 1
|_ servers: 0
|_ server: irc.Metasploitable.LAN
|_ version: Unreal3.3.3.1. irc.Metasploitable.LAN
|_ uptime: 0 days, 8:26:18
|_ source ident: nmap
|_ source host: 216f5410.00000000.ffff0044.IP
|_ error: Closing Link: dmhpkrtt[202.168.168.129] (Quit: dmhpkrtt)
8088/tcp open  ajp13    Apache/2.0.61 (Ubuntu)
|_ ajp-methods: Failed to get a valid response for the GET/300 request
8188/tcp open  http     Apache/2.0.61 (Ubuntu)
|_ http-server-header: Apache/2.0.61 (Ubuntu)
|_ http-title: Apache/2.0.61 (Ubuntu)
MAC Address: 08:00:C7:4D:E8:16 (VMware)
Device type: general purpose
Running: Linux 2.6.9
OS CPU: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.35
Network Distance: 1 hop
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPEs: cpe:/o:linux:linux_kernel

Host script results:
|_ smb-os-discovery:
|_ OS: Unix (Samba 3.0.2a-Debian)
|_ Computer name: metasploitable
|_ NetBIOS computer name:
|_ Domain name: localdomain
|_ FQDN: metasploitable.localdomain
|_ System time: 2016-02-07T16:29:33-0500
|_ smb-time: Protocol negotiation failed (SMB2)
|_ click-stream: none: 0x0000, deviation: 20.0000s, median: 0s
|_ smb-security-mode:
|_ account_used: blank
|_ authentication_level: user
|_ challenge_response: supported
|_ message_signing: disabled (dangerous, but default)
|_ hostinfo: NetBIOS name: METASPLOITABLE, NetBIOS user: unknown, NetBIOS MAC: unknown (unknown)

PROCESSES
TCP RTT      ADDRESS
1 1.98 ms 202.168.168.129 [202.168.168.129]

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 73.98 seconds
```

- sha256sum-nmap_scan-file.png (Integrity proof).

```
Session Acciones Editor Vista Ayuda
(kalilinux@KaliLinux)-[~]
$ sudo sha256sum nmap_scan.txt
1038b7c3e0e0268ed3c2b769d0492aacd19f6d8ed35f751252d587cbae731139  nmap_scan.txt
(kalilinux@KaliLinux)-[~]
$
```

Integrity Hash: nmap_scan.txt: 37804470200e008a687397e59670087702868019665557760773099908126830

Recommendations

Technical Action Plan:

- Decommission Insecure Services: Disable Telnet (Port 23) and vsftpd 2.3.4 (Port 21).
- Encryption: Transition all administrative traffic to SSH (Port 22).
- Update Cycle: Implement a patch management policy to update Apache and Samba versions, as the current versions are end-of-life (EOL).

Notes by tool

Nmap (v7.9x):

- Execution: `sudo nmap -sS -A -T4 192.168.106.129 -oN nmap_scan.txt`.
- Observation: The -A flag was highly effective for OS detection and script scanning, accurately identifying the vsftpd backdoor and several Samba vulnerabilities without manual scripts.

OpenSSL/Sha256sum:

- Observation: Vital for maintaining the chain of custody. The generated hash matches the one reported in the final Obsidian document.